

The Marshall Triangle: A Triadic Geometric Framework for Modeling Balance in Complex Systems

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Abstract

This paper introduces the Marshall Triangle, a formal geometric framework for representing balance among three competing system priorities. Inspired by color-space visualization, simplex geometry, and multi-objective tradeoff modeling, the framework maps triadic states into a continuous triangular domain in which relative dominance, pairwise blending, and equilibrium can be explored visually and mathematically. The model places primary influences at edge midpoints rather than vertices, producing a field in which secondary states emerge at the vertices and a white point marks neutral recombination within the rendered domain. A reference Harmony Index is defined from the normalized distance between a triadic state and balanced equilibrium, while the visualization layer is expressed as a Gaussian radial field over the triangular domain. The framework is implementation-independent but is illustrated through a reference rendering model using additive RGB contributions from midpoint sources. The Marshall Triangle is intended as a conceptual and computational tool for system design, visualization, and exploratory analysis in domains where competing priorities must be compared, calibrated, and communicated.

Keywords: triadic modeling; computational geometry; simplex visualization; system balance; harmony index; visual analytics; multi-objective tradeoffs; color-space models; radial basis functions

1. Introduction

Complex systems frequently require the reconciliation of competing priorities rather than the optimization of a single objective. In fields ranging from software architecture to interactive system design, practitioners must navigate tradeoffs among factors that cannot be simultaneously maximized. Conventional scalar metrics often obscure these tradeoffs, while high-dimensional models can become difficult to interpret or communicate.

The Marshall Triangle is proposed as a compact geometric framework for expressing triadic balance. Drawing inspiration from color-space visualization, simplex-based coordinate systems, and multi-objective tradeoff modeling, the framework encodes three competing axes within a triangular domain. This representation supports intuitive visualization of dominance, hybridization, and equilibrium, while remaining amenable to mathematical description and computational implementation.

The paper makes three contributions. First, it defines a midpoint-source triangular construction in which primary influences are positioned at edge midpoints rather than vertices. Second, it introduces a reference Harmony Index for quantifying deviation from triadic balance. Third, it describes a Gaussian radial-field rendering model that produces continuous visual gradients and a field-derived white point of neutral recombination. The result is a general-purpose visualization framework for exploring competing priorities in complex systems.

2. Related Work and Conceptual Context

The Marshall Triangle is related to, but distinct from, several established traditions. Colorimetry and color-space modeling provide a precedent for representing perceptual relationships through geometric domains and for treating white or neutral points as meaningful locations within a color

system [1–3]. Information visualization and visual analytics similarly emphasize the use of visual structures to support analytical reasoning, comparison, and interpretation [4–6].

The triangular domain also connects the framework to barycentric and simplex-based coordinate systems. In barycentric geometry, a point in a triangle may be represented by normalized weights relative to reference points, and these weights naturally express mixture, proportion, and relative contribution [7,8]. The Marshall Triangle adopts this general logic of normalized triadic representation, but its visual construction differs from a classical vertex-primary simplex by placing primary sources at edge midpoints.

The framework is also conceptually adjacent to multi-objective optimization, where tradeoffs arise because objectives often cannot be improved simultaneously without degrading others [9,10]. However, the Marshall Triangle is not presented as an optimization algorithm or decision rule. It is an interpretive geometric and visual model for representing relationships among competing priorities, making tradeoff structure easier to inspect, calibrate, and communicate.

Finally, the reference implementation uses radial Gaussian falloff, connecting the visualization layer to radial basis function methods, in which local influence is modeled as a distance-dependent field around source points [11,12]. This relation is conceptual rather than prescriptive: other interpolation or field models may be used if they preserve the core triadic interpretation.

3. Conceptual Overview

3.1. Triadic Balance

The Marshall Triangle is defined by three fundamental axes, each representing a competing priority within a system. In the reference application discussed throughout this paper, these axes correspond to Privacy, Performance, and Personalization. However, the framework itself is agnostic to the specific semantics of the axes; any triad of competing factors may be substituted.

In contrast to classical color triangles, the Marshall Triangle does not place primary axes at the vertices. Instead, the dominant source influences are located at the midpoints of each edge. The vertices therefore represent secondary dominance states emerging from the equal contribution of two axes, while interior points represent continuously blended states influenced by all three axes.

3.2. Emergent Equilibrium

The geometric center of the triangle corresponds to a symmetric balance among all three axes. However, the perceptual white point—the visual representation of neutral recombination—may shift based on calibration and weighting. This white point results from the recombination of triadic influences and emerges as a field-derived neutral point, even when the geometric center does not coincide with calibrated perceptual balance.

4. Geometric Construction

The Marshall Triangle is constructed as an equilateral triangle embedded in a two-dimensional plane. Its defining innovation lies in relocating the primary source influences from the vertices—typical of classical color-triangle models—to the midpoints of each edge. Each midpoint anchors one of the three fundamental axes: Privacy, Performance, or Personalization. This inversion of conventional triangle logic establishes a continuous field of influence across the interior, rather than partitioned zones of dominance.

Within this configuration, each point in the triangular domain is associated with non-negative source influences that may be normalized to express the relative contributions of the three axes. The vertices emerge not as sources, but as extrema—locations where the contribution from two adjacent sources dominates while the third is minimized. This geometry creates a bounded, interpretable domain in which perceptual gradients, patterns of relative dominance, and emergent equilibrium states can be visualized and calibrated with mathematical rigor.



Figure 1. The Marshall Triangle in its canonical visual form, showing a continuous triadic field structured by midpoint-based primary influences and an emergent white point of neutral recombination.

5. Mathematical Formulation

Let a triadic system state be represented by a vector $s = (s_1, s_2, s_3)$, where each component is non-negative and at least one component is positive. The normalized state vector is

$$p_i = \frac{s_i}{\sum_{j=1}^3 s_j}, \quad i \in \{1, 2, 3\}. \quad (1)$$

The balanced reference state is $u = (1/3, 1/3, 1/3)$. A reference Harmony Index may be defined as

$$H(s) = 1 - \frac{\|p - u\|_2}{\sqrt{2/3}}, \quad (2)$$

where $\sqrt{2/3}$ is the Euclidean distance from the balanced state to any vertex of the probability simplex and therefore the maximum attainable value of $\|p - u\|_2$. Under this definition, $H \in [0, 1]$, with $H = 1$ at perfect triadic balance and $H = 0$ at single-axis dominance.

For visualization, let Ω denote the triangular domain and let $c_1, c_2, c_3 \in \Omega$ denote the three edge-midpoint source locations. Distances on Ω are Euclidean distances in the planar embedding of the equilateral triangle, with all coordinates expressed in the same normalized units. For a point $x \in \Omega$, the radial influence of source i may be written as

$$G_i(x) = \exp\left(-\frac{\|x - c_i\|^2}{2\sigma^2}\right). \quad (3)$$

The parameter $\sigma > 0$ controls the radial falloff width and may be chosen as a calibrated fraction of the normalized triangle scale. Let e_1, e_2, e_3 denote the red, green, and blue basis channels. A rendered color field can then be expressed as

$$C(x; s) = N \left(\sum_{i=1}^3 p_i G_i(x) e_i \right), \quad (4)$$

where $N(\cdot)$ is a hue-preserving normalization operator, for example $N(v) = v / \max(\|v\|_\infty, \epsilon)$ for a small $\epsilon > 0$, or an equivalent max-value normalization used to avoid RGB clipping. The reference implementation uses max-value normalization, but the framework does not require a single interpolation method. Valid implementations may employ barycentric coordinates, radial falloff fields, or other continuous blending strategies consistent with triadic equilibrium.

6. Illustrative Examples

Figure 1 presents the Marshall Triangle in its canonical visual form. Figure 2 identifies the primary and pairwise regions of the triangle, distinguishing midpoint-based primary influences from vertex-adjacent blended regions. Figure 3 then compares three representative system states: a dual-axis emphasis state, a balanced default state, and a single-axis Privacy-emphasis state.

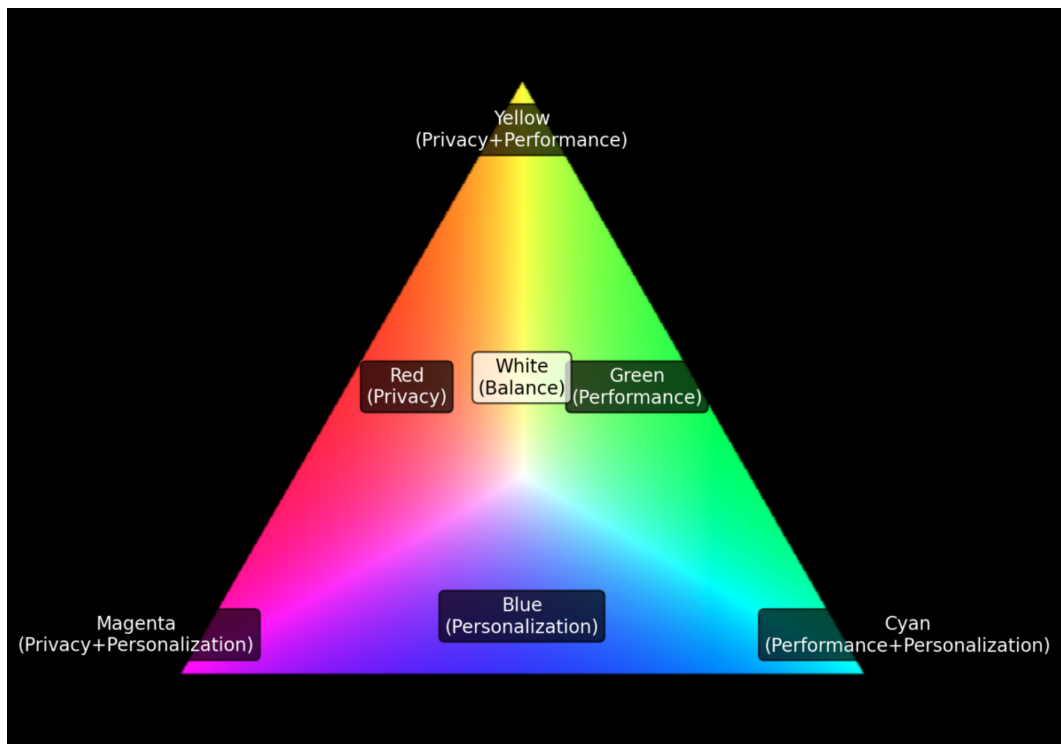


Figure 2. Labeled structural interpretation of the Marshall Triangle. Primary influences are positioned at edge midpoints, while vertex-adjacent regions express pairwise blending between adjacent axes. The interior field represents continuously blended triadic states, with the white point marking computed neutral recombination within the field.

Across these examples, the white point should not be read as a pointer toward the dominant axis. The white point is field-derived rather than axis-indicative: it marks the computed location of neutral recombination within the resulting color field. When one axis expands, the white point may shift away from that axis even as the corresponding field becomes more visually dominant across the triangle. When two axes expand together, the white point may shift toward the relatively minimized third axis. In both cases, the movement reflects the recalculated field geometry rather than a separate compensatory mechanism.

This construction also distinguishes primary-axis emphasis from pairwise blending. Near each edge midpoint, the corresponding primary axis is emphasized; near the vertices, adjacent axes blend into secondary dominance regions. The interior of the triangle represents continuously blended

triadic states, where smooth gradients reflect varying relationships among Privacy, Performance, and Personalization.

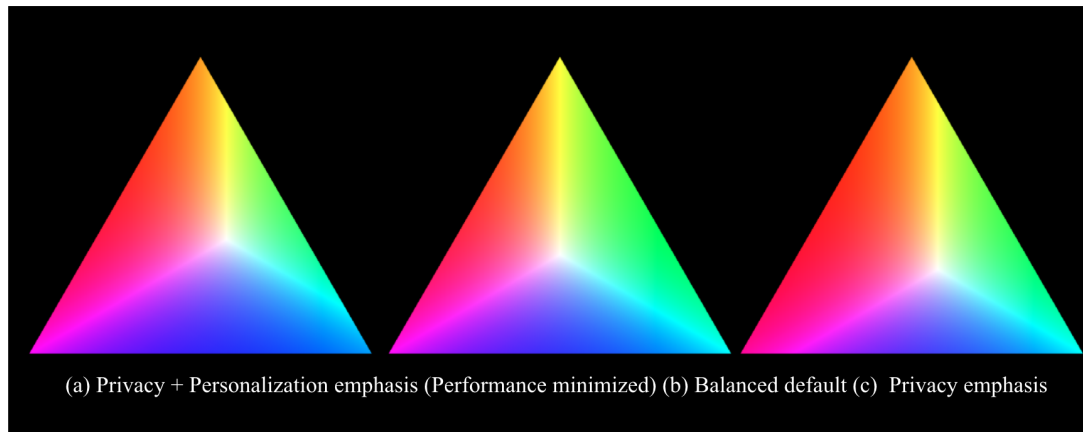


Figure 3. Representative Marshall Triangle state comparison. (a) Privacy + Personalization emphasis, with Performance comparatively minimized. (b) Balanced default state. (c) Privacy-emphasis state. These panels illustrate that the white point is field-derived rather than axis-indicative: it marks the computed location of neutral recombination within the resulting color field, not the location of the dominant influence.

These examples are intended to demonstrate the expressive behavior of the framework rather than prescribe a single computational implementation. Small adjustments to triadic weighting produce coherent visual transitions, enabling calibration, interpretation, and comparison of system states within a shared geometric field.

7. Discussion

The Marshall Triangle offers a middle ground between oversimplified scalar metrics and unwieldy high-dimensional models. By constraining analysis to a triadic domain, it foregrounds tradeoffs and balance in a visually and mathematically tractable form.

The framework is not intended to replace formal multi-objective optimization, empirical user modeling, or domain-specific decision analysis. Instead, it provides a compact representational layer for exploring and communicating triadic relationships before, during, or after more specialized analysis. Its interpretive value lies in making competing priorities visible as continuous fields rather than isolated scores.

The reference implementation adopts radial Gaussian falloff calibrated to an emergent white point. This implementation choice supports smooth visual transitions and intuitive manipulation, but the underlying framework remains broader than any single software realization. Future implementations may adapt the interpolation method, calibration procedure, or semantic axes while preserving the central midpoint-source construction.

8. Limitations and Future Work

The framework is intentionally formal and conceptual. It does not by itself establish empirical validity, normative priority weights, or domain-specific decision authority. The Privacy, Performance, and Personalization axes used in the reference application are illustrative rather than exhaustive.

Future work may include empirical studies of interpretability, formal comparison with barycentric and radial basis function interpolation methods, applications to additional triadic domains, and integration with decision-support systems. Extensions to higher-order simplices may also allow analogous visualization of four or more competing priorities, though such extensions would sacrifice some of the visual immediacy of the triangular form.

9. Conclusions

The Marshall Triangle provides a concise geometric framework for reasoning about balance among three competing priorities. By combining interpretability, mathematical structure, and visual expressiveness, it offers a flexible conceptual tool for analyzing complex systems. Its abstraction from specific implementations allows it to serve as a unifying model across domains where triadic tradeoffs are fundamental.

Author Contributions: Conceptualization, software, visualization, formal analysis, and manuscript preparation were conducted solely by the author, Paul W. Marshall.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The source code, figures, and manuscript source files are available through the project repository at <https://github.com/Paul-W-Marshall/marshall-triangle> and are archived at <https://doi.org/10.5281/zenodo.20696528>. The interactive implementation is available at <https://marshalltriangle.app>.

Acknowledgments: The author acknowledges the use of OpenAI ChatGPT and Anthropic Claude for editorial assistance, grammar correction, formatting support, and version tracking. The author reviewed and approved all content and takes full responsibility for the manuscript.

Conflicts of Interest: The author is the originator of the Marshall Triangle framework and maintains related public project materials and software implementations. The author is the named inventor on U.S. Utility Patent Application No. 19/734,975, which concerns related system architecture and implementations. No external funding or third-party financial conflicts are declared.

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